PROJECT REPORT · 221601907



HeartSense AI

An AI-Powered Heart Disease Prediction

Capstone Project – II · System Design and Implementation

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PROGRAMME

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1.1 · BRIEF OF SYSTEM

Introduction

HeartSense AI is a machine learning-based web application that estimates the likelihood of heart disease from patient health data.

Heart disease is one of the major health concerns worldwide. Early identification of risk factors and timely medical consultation can help improve patient care.

The main objective is an accessible, data-driven decision-support tool that helps users and healthcare professionals understand potential heart disease risk.

Clinical inputs

Age, sex, chest pain type, resting blood pressure, cholesterol level, resting ECG results, maximum heart rate and other relevant clinical features.

Machine learning models

Logistic Regression, Random Forest, Support Vector Machine (SVM) and XGBoost are trained and compared, with SHAP or LIME explanations where supported.

Application features

User authentication, role-based access for patients, doctors and administrators, patient data management, risk assessment, health education recommendations, interactive dashboards and PDF reports.

Intended for educational and research purposes. It does not replace professional medical advice, clinical diagnosis or treatment.

1.2 · BRIEF OF SYSTEM

Problem Statement

The need for an accessible, data-driven system that analyzes patient health parameters and estimates the likelihood of heart disease.

Traditional assessment

Relies on clinical examinations, medical history and diagnostic tests that need professional expertise to interpret.

Manual interpretation

Reading many medical attributes is time-consuming, and patients may find it hard to understand the factors behind their risk.

Proposed system

Logistic Regression, Random Forest, SVM and XGBoost compare prediction performance, with explanations, risk levels and general preventive recommendations.

A web-based prediction and clinical decision-support system with secure patient data management, explainable predictions, interactive dashboards and PDF report generation.

Supports educational and research activities and assists clinical review; it does not replace professional diagnosis or medical decision-making.

1.3 · OBJECTIVES 1–8

Objective of the System

1 Patient Data Collection

Enter age, sex, chest pain type, blood pressure, cholesterol, ECG results and maximum heart rate.

2 Secure User Authentication

Secure registration and login with role-based access for patients, doctors and administrators.

3 Patient Data Management

Authorized users add, update, view and manage patient records and history securely.

4 Data Preprocessing

Handle missing values, remove duplicates, encode categories and normalize numerical data.

5 Machine Learning Model Training

Train and evaluate Logistic Regression, Random Forest, SVM and XGBoost.

6 Model Performance Comparison

Compare accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.

7 Heart Disease Prediction

Analyze patient health parameters to estimate the likelihood of heart disease.

8 Explainable AI (XAI)

Explain predictions with SHAP or LIME and identify the contributing health features.

1.3 · OBJECTIVES 9–16

Objective of the System

9 Risk Assessment

Categorize results as Low, Moderate or High Risk using documented thresholds and probability.

10 Personalized Recommendations

General health education, lifestyle, diet and exercise guidance, plus advice to seek medical consultation.

11 Dashboard and Visualization

Show results, patient statistics, parameter charts, history and model metrics interactively.

12 PDF Report Generation

Downloadable, printable reports with details, results, risk, explanations and recommendations.

13 Prediction History Management

Keep prediction records so authorized users can review results and changes over time.

14 Administrative Management

Manage user accounts, roles, system records and application activities by permission.

15 Data Security and Reliability

Protect sensitive data, enforce access controls and keep inputs and records correct.

16 Clinical Decision Support

Support understanding of heart disease risk; predictions are estimates, not a diagnosis.

BRIEF OF SYSTEM

Machine Learning Approach

1

Preprocess

Handle missing values, remove duplicates, encode categories and normalize numbers.

2

Train

Train and evaluate four machine learning algorithms.

3

Compare

Accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.

4

Predict

Estimate heart disease likelihood and show the probability where supported.

5

Explain

SHAP or LIME feature contributions, risk level and recommendations.

Algorithms compared

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Risk levels

- Low Risk
- Moderate Risk
- High Risk

1.4 · BRIEF OF SYSTEM

Hardware and Software Requirements

Hardware

Processor

Intel Core i3 or equivalent and above

RAM

Minimum 4 GB; 8 GB or more recommended

Storage

Sufficient free space for the dataset, patient records, trained machine learning models and generated PDF reports

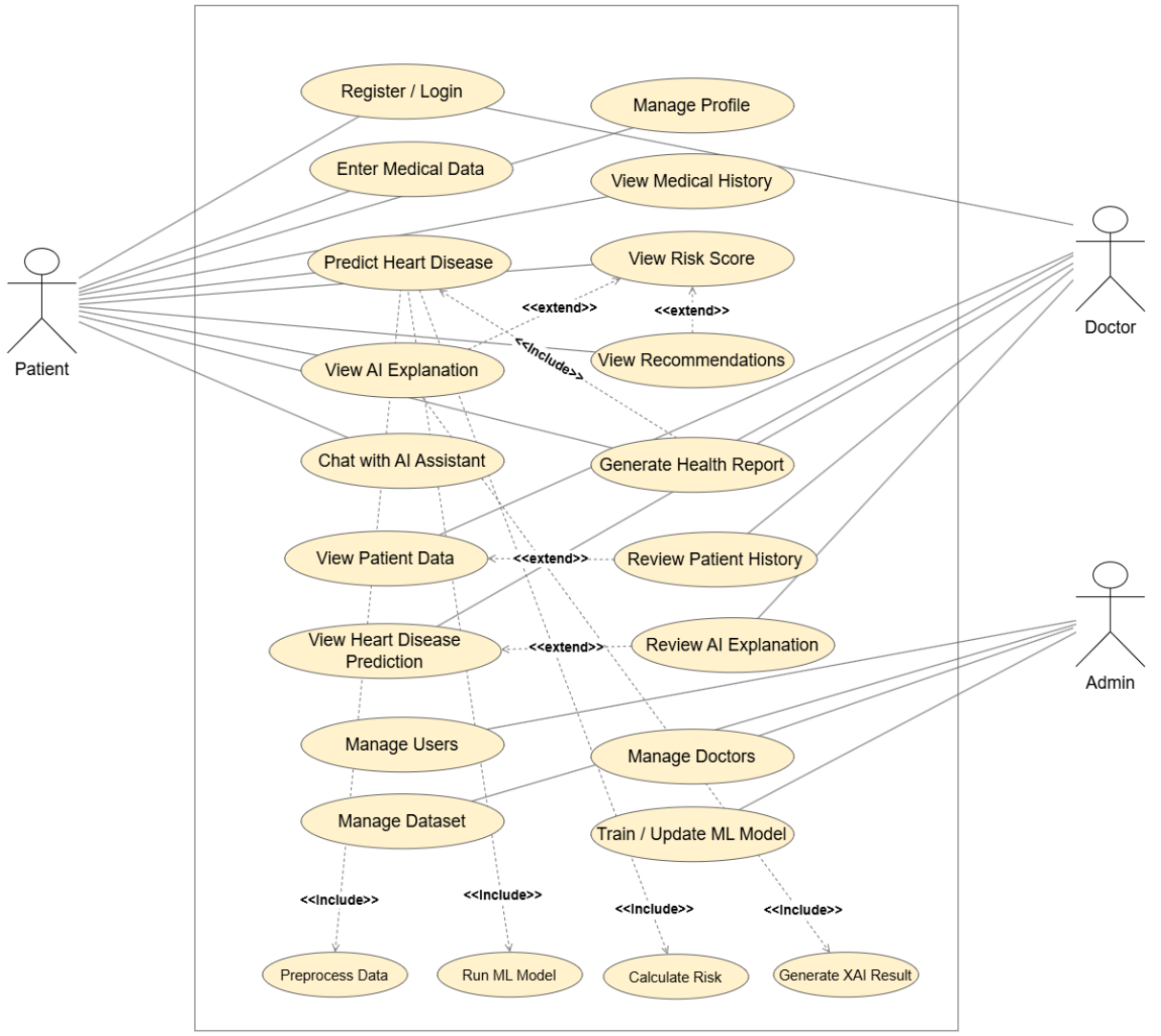
Software

Component	Technology
Operating System	Windows 10/11 or a compatible operating system
Frontend	React.js, HTML, CSS and JavaScript
Backend	Python and Django REST Framework
Database	SQLite for development; PostgreSQL can be used for deployment
Machine Learning	Scikit-learn, Pandas, NumPy and XGBoost
Explainable AI	SHAP or LIME
Data Visualization	Recharts or Chart.js
PDF Generation	ReportLab
Data Processing	Pandas and NumPy
Development Tools	Visual Studio Code
Browser	Google Chrome or Microsoft Edge

2.1 · UML DIAGRAM

Use-Case Diagram

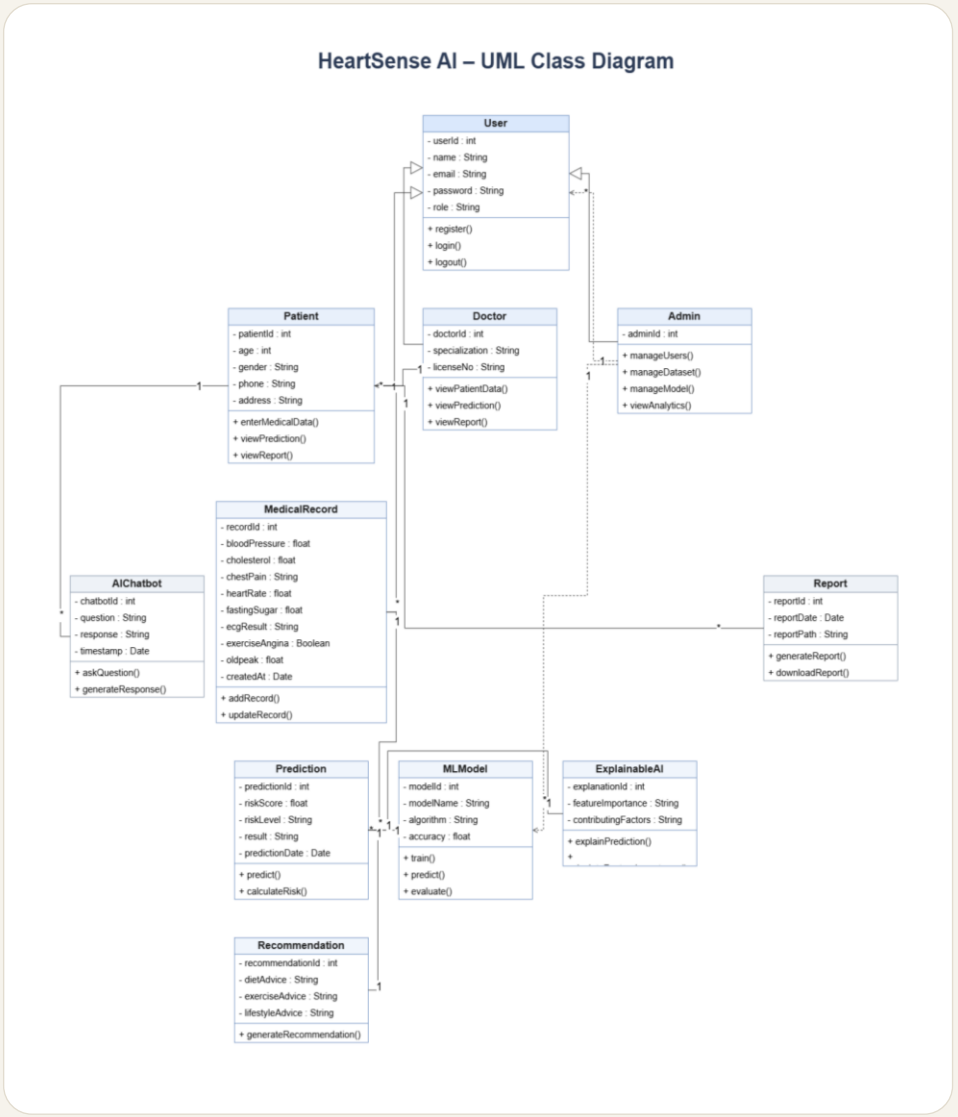
Actors and their interactions with HeartSense AI across the patient, doctor and administrator roles.



2.2 · UML DIAGRAM

Class Diagram

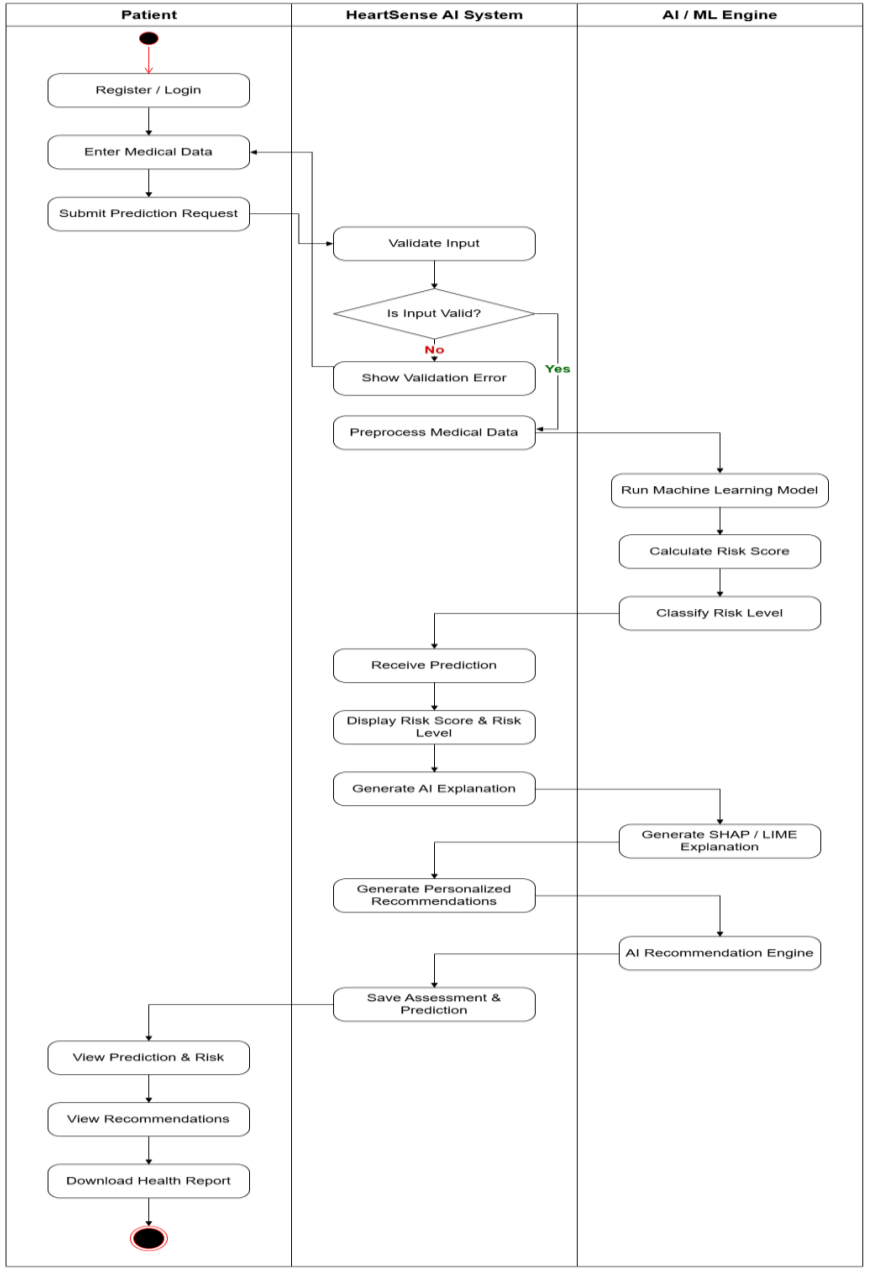
Classes, attributes and relationships behind the HeartSense AI data model.



2.3 · UML DIAGRAM

Activity Diagram

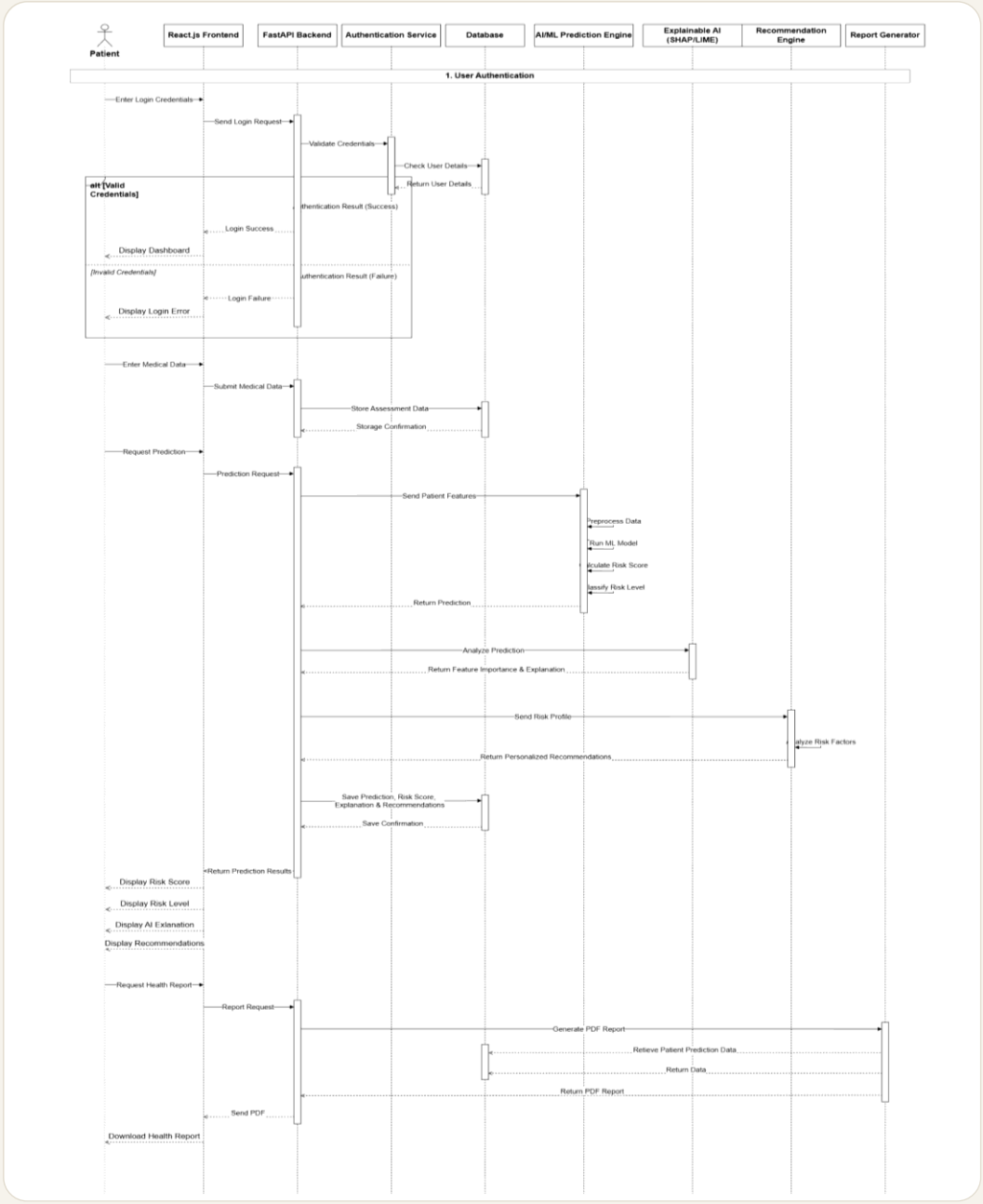
The flow of activities in HeartSense AI from start to finish.



2.4 · UML DIAGRAM

Sequence Diagram

The order of interactions between the parts of HeartSense AI.



3 · DATA DICTIONARY

Ten Database Tables

3.1 User
Accounts, roles and status

3.2 Patient
Profile and baseline health data

3.3 Medical Assessment
Clinical inputs for each assessment

3.4 Prediction
Model output, risk score and level

3.5 Explainable AI
SHAP or LIME feature contributions

3.6 Recommendation
Diet, exercise, lifestyle and medical advice

3.7 AI Chat
User questions and AI responses

3.8 Report
Generated PDF report records

3.9 ML Model
Algorithms and evaluation metrics

3.10 Admin
Permissions and admin sign-ins

Foreign keys link User → Patient → Medical Assessment → Prediction, which in turn feeds Explainable AI, Recommendation and Report.

3.1 · DATA DICTIONARY

User Table

Field Name	Data Type	Key	Description
User_id	INT	PK	Unique user identifier
Full_name	VARCHAR	—	User's full name
Email	VARCHAR	UNIQUE	User email address
Password	VARCHAR	—	Encrypted password hash
Role	ENUM	—	User role: Patient, Doctor or Admin
Phone	VARCHAR	—	Contact phone number
Created_at	DATETIME	—	Account creation timestamp
Status	BOOLEAN	—	Account status (Active/Inactive)

3.2 · DATA DICTIONARY

Patient Table

Field Name	Data Type	Key	Description
Patient_id	INT	PK	Unique patient identifier
User_id	INT	FK	References User(user_id)
Age	INT	—	Patient age in years
Gender	VARCHAR	—	Gender (Male / Female / Other)
Height	DECIMAL	—	Height in centimeters
Weight	DECIMAL	—	Weight in kilograms
Blood_pressure	VARCHAR	—	Baseline blood pressure reading
Cholesterol	FLOAT	—	Cholesterol level
Fasting_blood_sugar	FLOAT	—	Fasting blood sugar level
Smoking_status	BOOLEAN	—	Smoking history flag
Family_history	BOOLEAN	—	Family history of heart disease
Created_at	DATETIME	—	Record creation timestamp

3.3 · DATA DICTIONARY

Medical Assessment Table

Field Name	Data Type	Key	Description
Assessment_id	INT	PK	Unique assessment identifier
Patient_id	INT	FK	References Patient(patient_id)
Chest_pain_type	VARCHAR	—	Type of chest pain experienced
Resting_bp	FLOAT	—	Resting blood pressure (mmHg)
Cholesterol	FLOAT	—	Serum cholesterol (mg/dl)
Fasting_bs	BOOLEAN	—	Fasting blood sugar > 120 mg/dl
Resting_ecg	VARCHAR	—	Resting ECG results
Max_heart_rate	INT	—	Maximum heart rate achieved
Exercise_angina	BOOLEAN	—	Exercise-induced angina status
Oldpeak	FLOAT	—	ST depression induced by exercise
St_slope	VARCHAR	—	Slope of peak exercise ST segment
Vessels	INT	—	Number of major vessels colored by fluoroscopy
Thalassemia	VARCHAR	—	Thalassemia diagnostic result
Assessment_date	DATETIME	—	Date and time of assessment

3.4 · DATA DICTIONARY

Prediction Table

Field Name	Data Type	Key	Description
Prediction_id	INT	PK	Unique prediction identifier
Assessment_id	INT	FK	References Medical Assessment(assessment_id)
Model_name	VARCHAR	—	ML model used for evaluation
Prediction	VARCHAR	—	Predicted disease outcome
Risk_score	FLOAT	—	Calculated risk percentage (0-100%)
Risk_level	VARCHAR	—	Risk classification (Low, Moderate, High)
Confidence_score	FLOAT	—	Model confidence score
Prediction_date	DATETIME	—	Timestamp of prediction generation

3.5 · DATA DICTIONARY

Explainable AI Table

Field Name	Data Type	Key	Description
Explanation_id	INT	PK	Unique explanation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Method	VARCHAR	—	XAI method used (SHAP / LIME)
Feature_name	VARCHAR	—	Key medical feature evaluated
Feature_value	VARCHAR	—	Patient's feature value
Contribution	FLOAT	—	Feature contribution score to prediction
Explanation_text	TEXT	—	Human-readable explanation text

3.6 · DATA DICTIONARY

Recommendation Table

Field Name	Data Type	Key	Description
Recommendation_id	INT	PK	Unique recommendation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Diet_advice	TEXT	—	General dietary recommendations
Exercise_advice	TEXT	—	Physical activity and exercise guidelines
Lifestyle_advice	TEXT	—	General lifestyle modifications
Medical_advice	TEXT	—	Clinical recommendations & specialist care guidance
Created_at	DATETIME	—	Timestamp of recommendation creation

3.7 · DATA DICTIONARY

AI Chat Table

Field Name	Data Type	Key	Description
Chat_id	INT	PK	Unique chat log identifier
User_id	INT	FK	References User(user_id)
Question	TEXT	—	User's query/question text
Ai_response	TEXT	—	AI-generated conversational response
Created_at	DATETIME	—	Chat interaction timestamp

3.8 · DATA DICTIONARY

Report Table

Field Name	Data Type	Key	Description
Report_id	INT	PK	Unique report identifier
Patient_id	INT	FK	References Patient(patient_id)
Prediction_id	INT	FK	References Prediction(prediction_id)
Report_file	VARCHAR	—	PDF report file path or filename
Generated_by	INT	FK	References User(user_id) who generated report
Generated_at	DATETIME	—	Report generation timestamp

3.9 · DATA DICTIONARY

ML Model Table

Field Name	Data Type	Key	Description
Model_id	INT	PK	Unique model identifier
Model_name	VARCHAR	—	Model name/identifier
Algorithm	VARCHAR	—	Algorithm (Random Forest, SVM, XGBoost, etc.)
Accuracy	FLOAT	—	Model accuracy score
Precision	FLOAT	—	Model precision score
Recall	FLOAT	—	Model recall score
F1_score	FLOAT	—	F1 score
Auc_score	FLOAT	—	ROC-AUC evaluation score
Version	VARCHAR	—	Model version string
Trained_date	DATETIME	—	Model training completion date
Status	BOOLEAN	—	Model active status flag

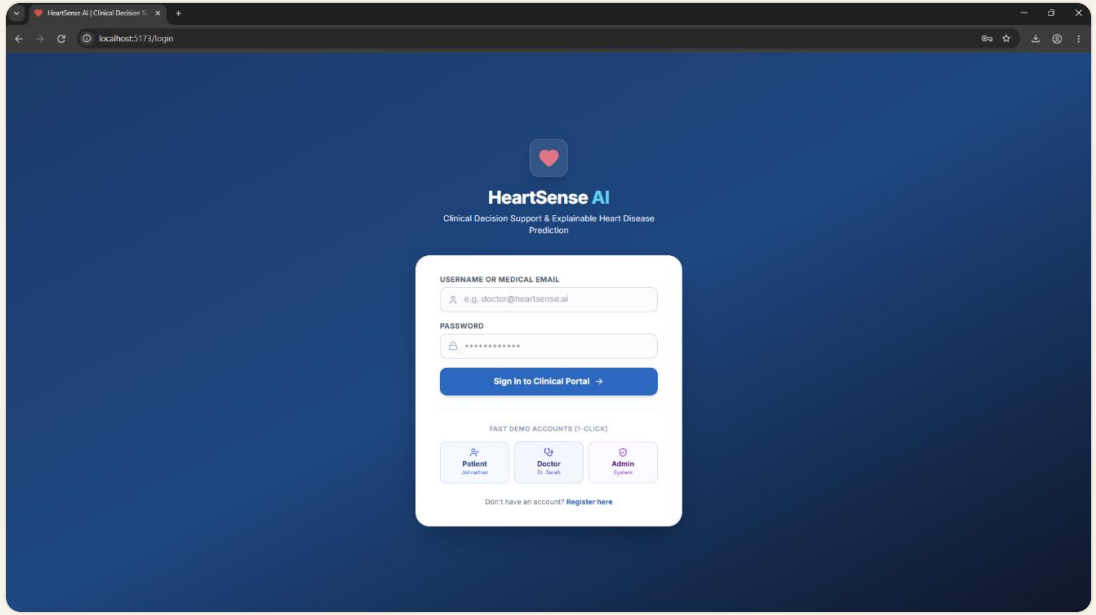
3.10 · DATA DICTIONARY

Admin Table

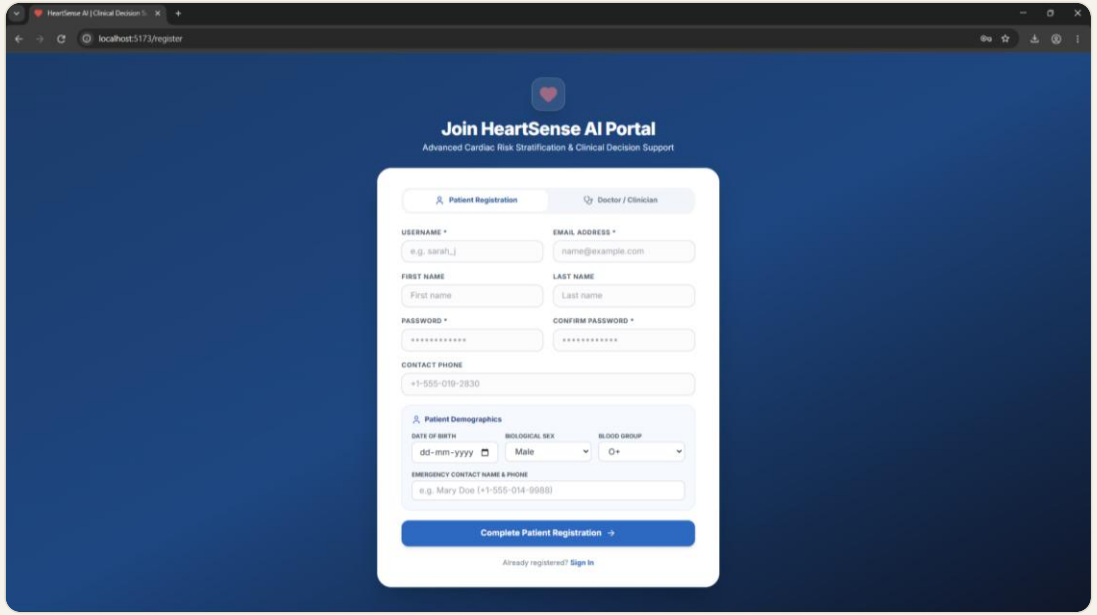
Field Name	Data Type	Key	Description
Admin_id	INT	PK	Unique administrator identifier
User_id	INT	FK	References User(user_id)
Permissions	TEXT	—	JSON/Text specification of permissions
Last_login	DATETIME	—	Last administrative login timestamp
Status	BOOLEAN	—	Admin account active status

4 · SCREENSHOTS

Login and Register



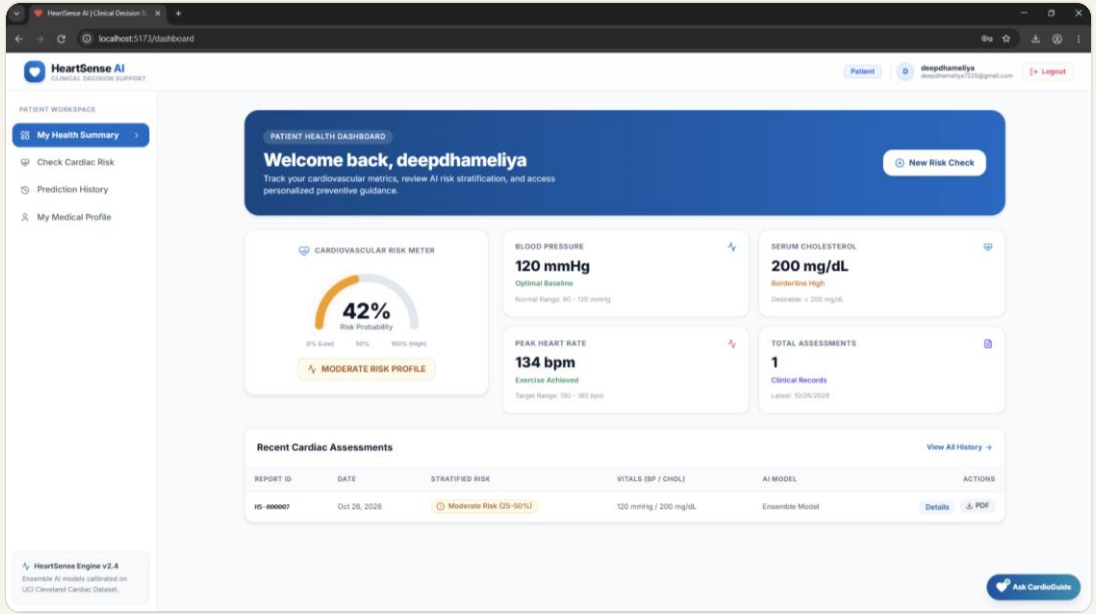
4.1 Login Page



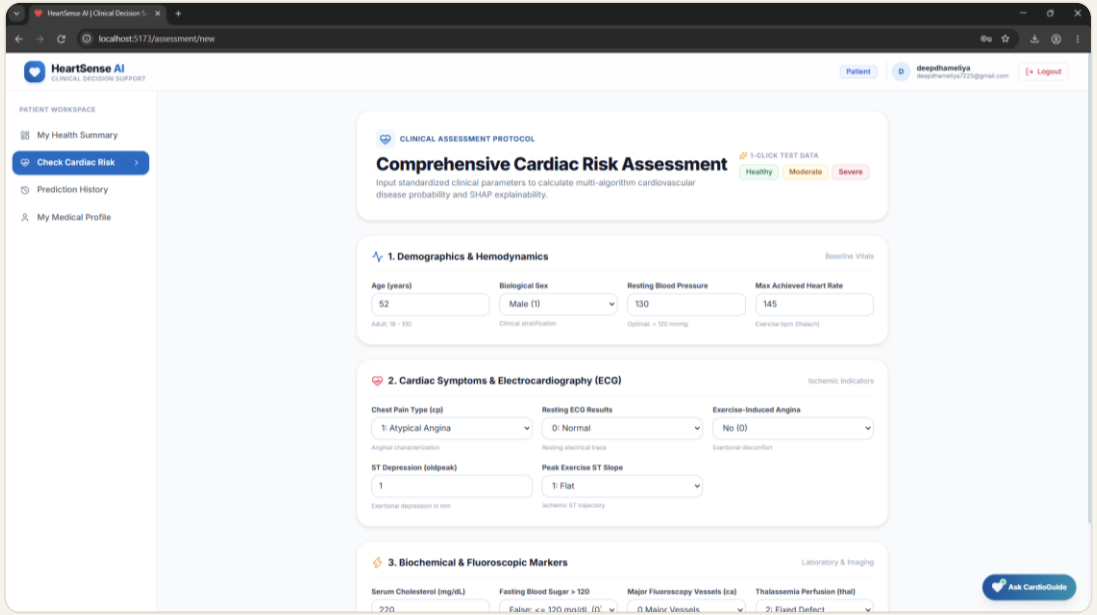
4.2 Register Page

4.3 · PATIENT SIDE

Dashboard and Risk Check



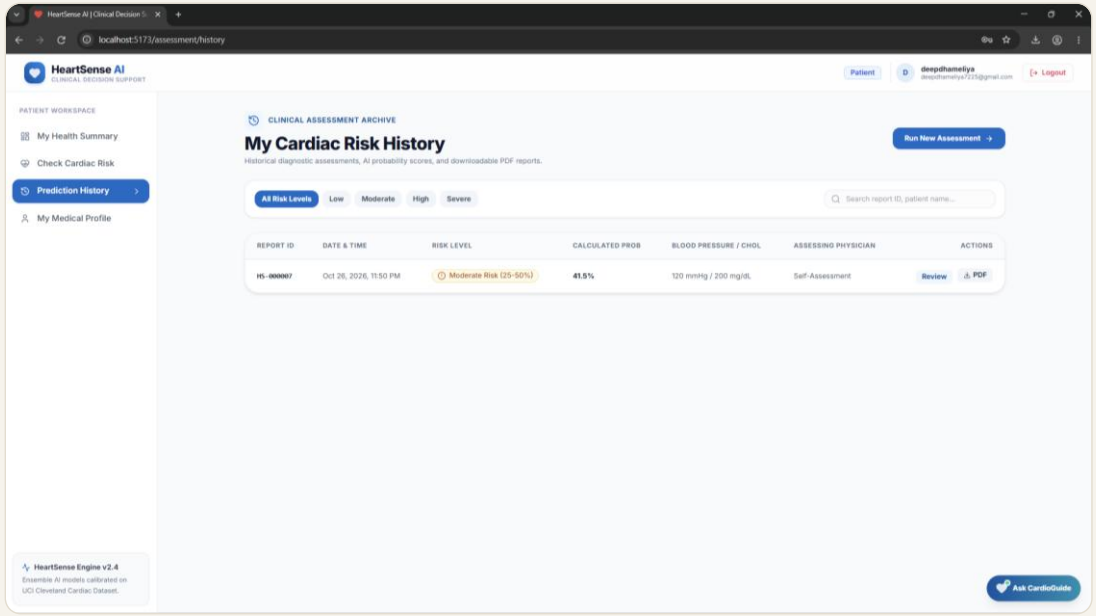
4.3.1 Dashboard Page



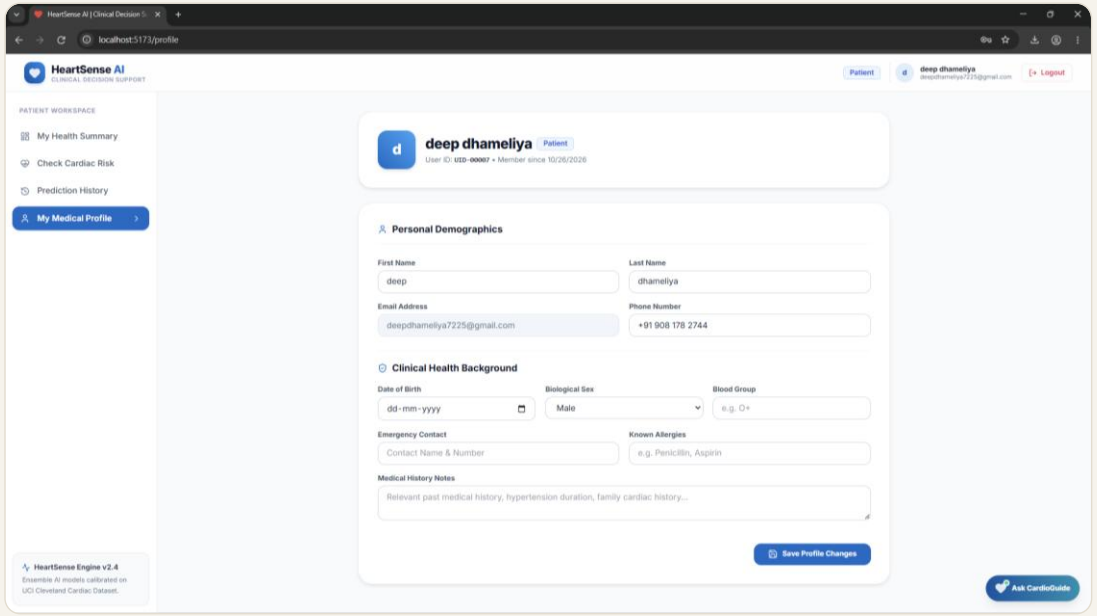
4.3.2 Risk Check Page

4.3 · PATIENT SIDE

History and Profile



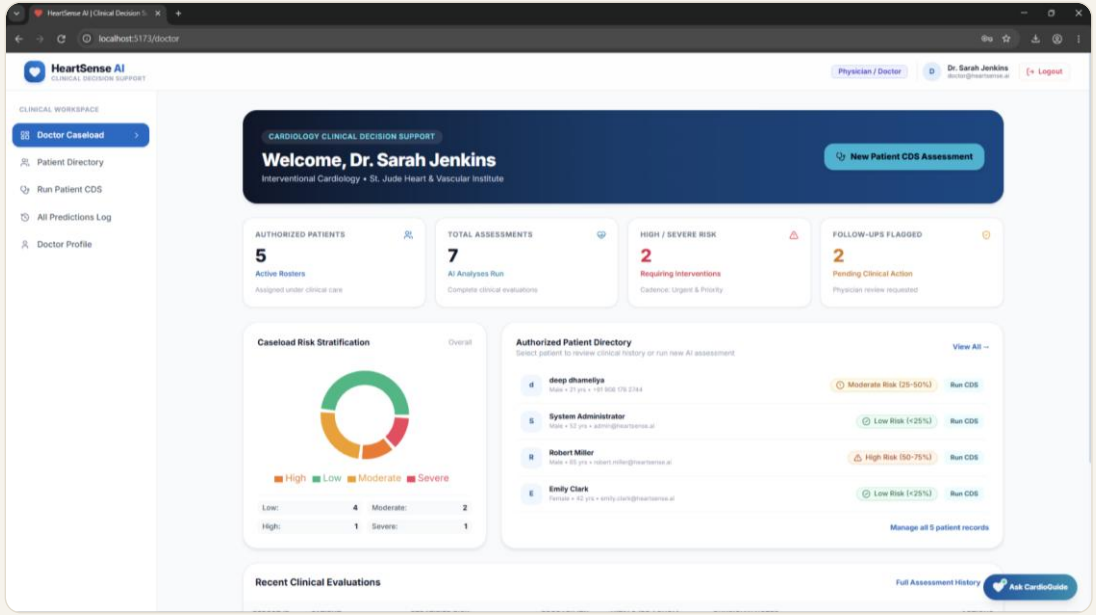
4.3.3 History Page



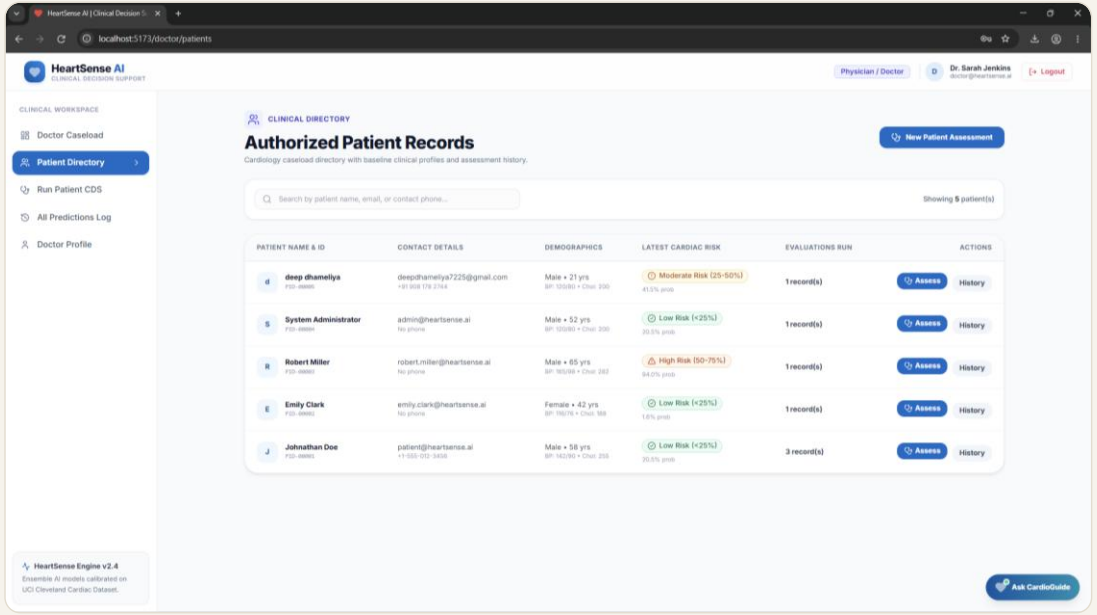
4.3.4 Profile Page

4.4 · DOCTOR SIDE

Dashboard and Patients Directory



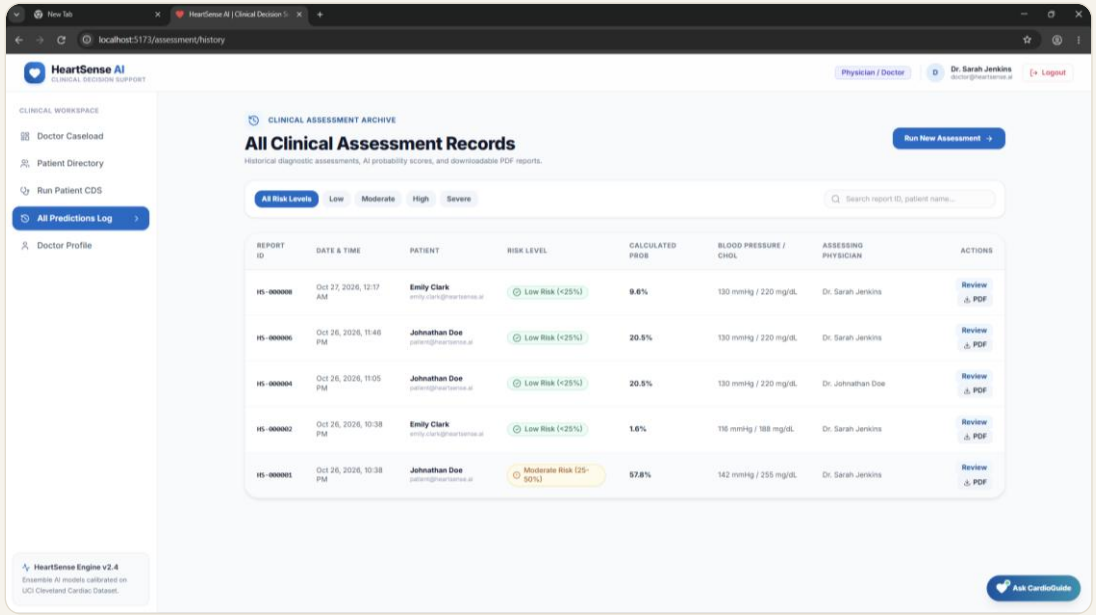
4.4.1 Dashboard Page



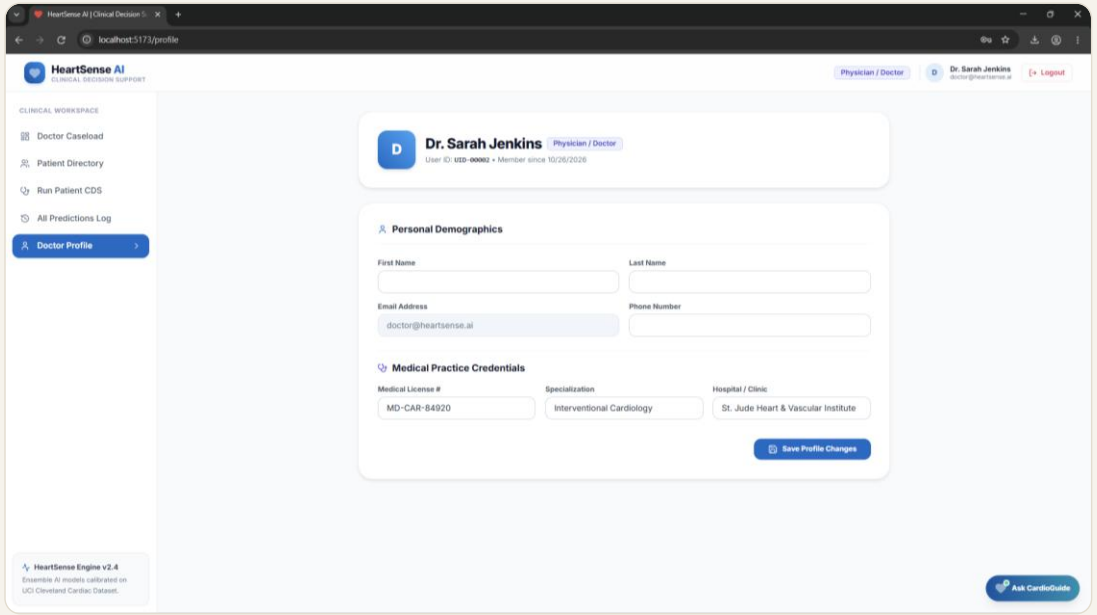
4.4.2 Patients Directory Page

4.4 · DOCTOR SIDE

Clinical Record and Profile



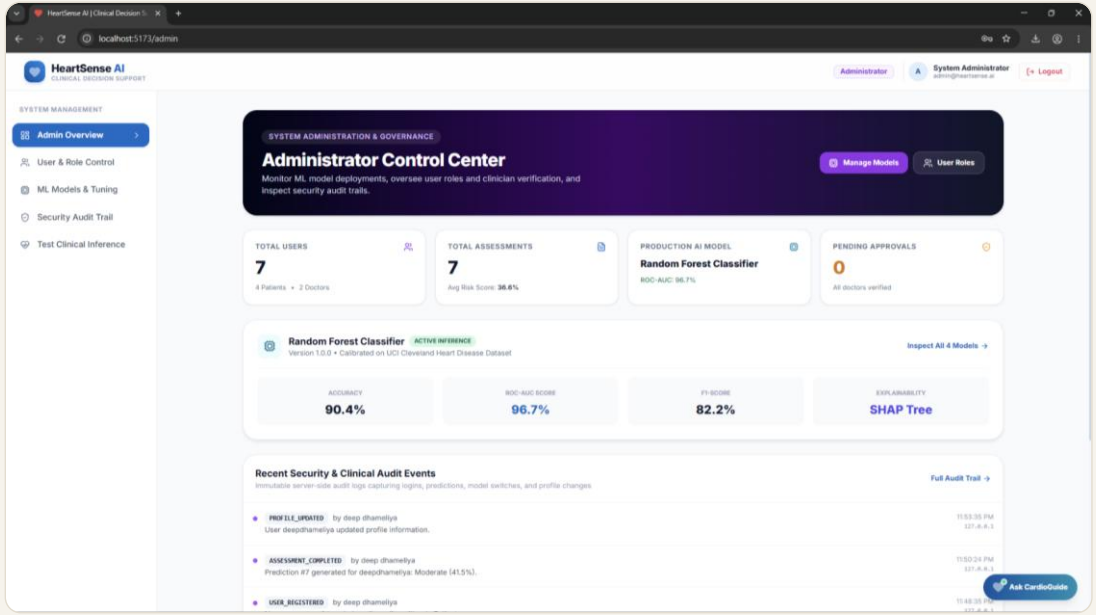
4.4.3 Clinical Record Page



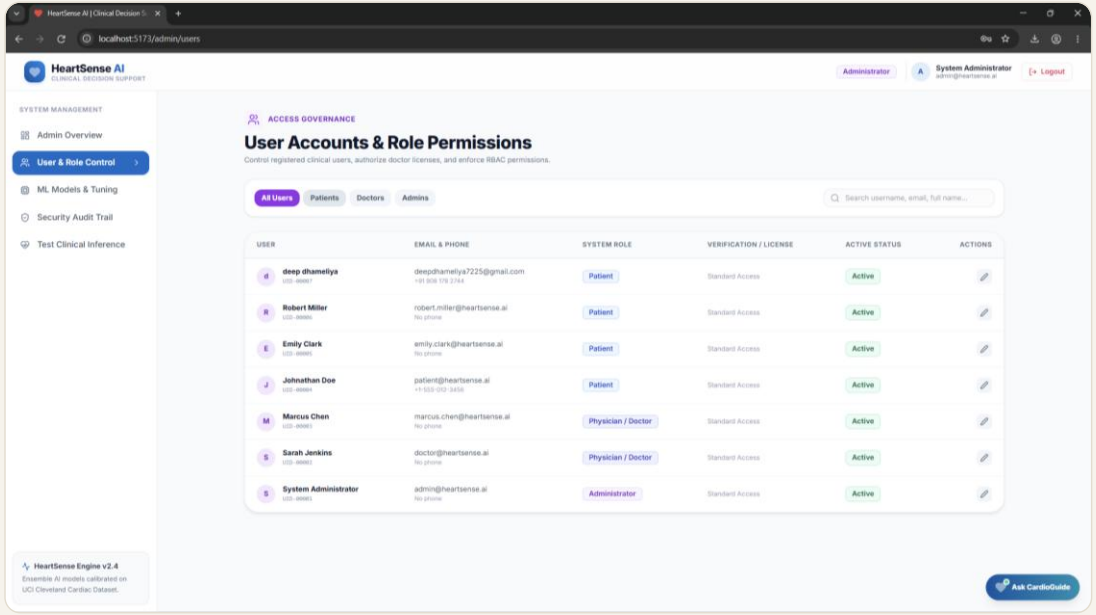
4.4.4 Profile Page

4.5 · ADMIN SIDE

Dashboard and User Management



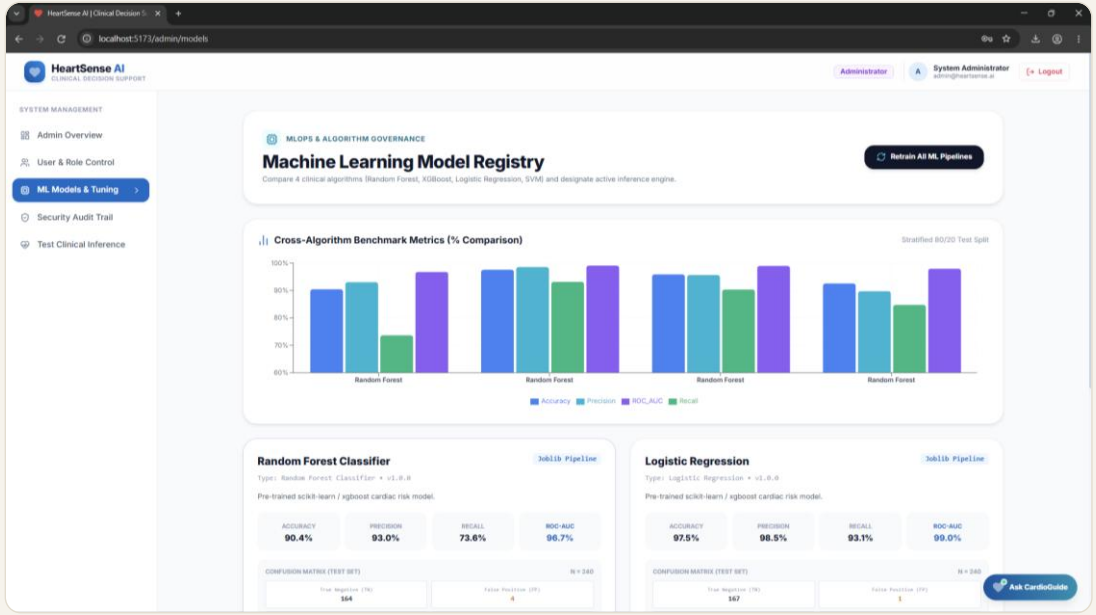
4.5.1 Dashboard Page



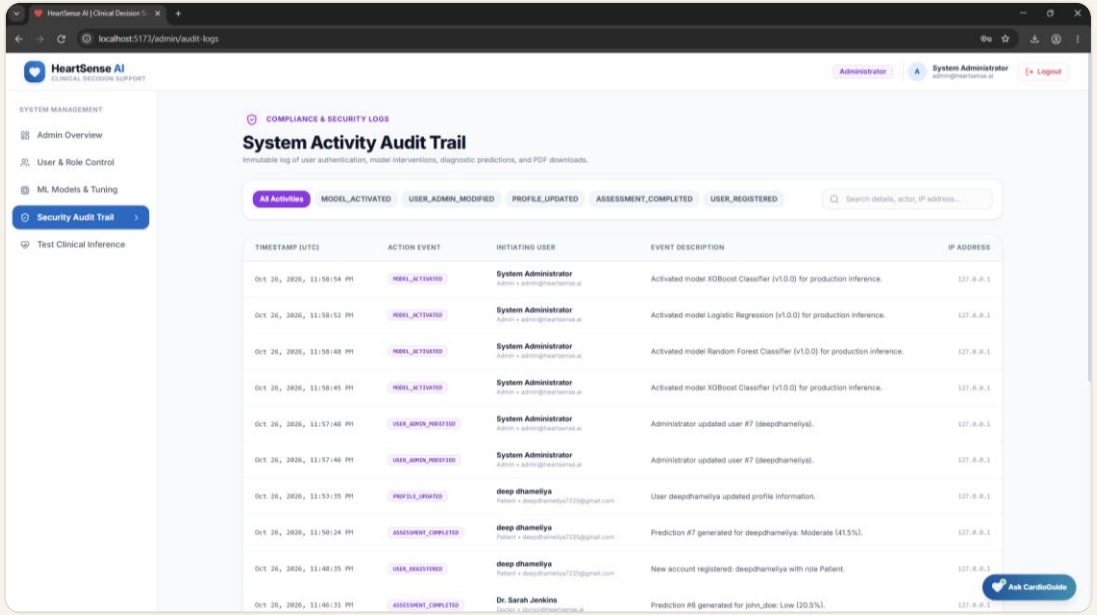
4.5.2 User Management Page

4.5 · ADMIN SIDE

Model Tuning and Audit Logs



4.5.3 Model Tuning Page



4.5.4 Audit Logs Page

References

Resource	Link
UCI Heart Disease dataset	archive.ics.uci.edu/dataset/45/heart+disease
Django documentation	docs.djangoproject.com/en/6.1/
React documentation	react.dev
XGBoost documentation	xgboost.readthedocs.io/en/stable/
SHAP documentation	shap.readthedocs.io/en/latest/
ReportLab documentation	docs.reportlab.com
Python documentation	docs.python.org/3/

HEARTSENSE AI



Thank You

An AI-Powered Heart Disease Prediction

Dhameliya Deep · Pansuriya Mihir · Guide: Dr. Jyoti Dubey

HeartSense AI is intended for educational and research purposes and does not replace professional medical advice, clinical diagnosis or treatment.